Carnegie Mellon University
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Thesis Proposal

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Advances in Time Series Analysis: Simulation-based Approaches to Estimation, Testing for Structure, and Prediction with Transformers

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Abstract

Technological advances have enabled the collection of large-scale time series data in many fields. Traditional time series methods often struggle to capture the complex dynamics present in such data because of their restrictive assumptions, such as stationarity, linear dynamics, or Gaussian noise. Consequently, there is growing demand for more flexible time series methods. At the same time, recent developments in machine learning, particularly the transformer architecture, have demonstrated remarkable success in sequence modeling. However, the statistical foundations of transformer-based methods for time series, particularly in nonstationary settings, remain poorly understood.

In this thesis proposal, we present simulation-based methods for problems in multivariate time series analysis. In the first part, we introduce a bootstrap-based framework for conditional independence testing with nonstationary time series that utilizes nonlinear regression. In the second part, we develop a simulation-based parameter estimation framework for time series models, including nonlinear non-Gaussian state-space models, discrete-time dynamical systems and ODEs with observational noise, and discretized Lévy-driven SDEs. In the third part, we introduce a simulation-and-regression framework for time series prediction with transformers.

Contents

1 Introduction

1.1 Thesis Agenda

We provide a concise overview of the organization of this thesis proposal, along with some motivations for each topic.

I. Conditional Independence Testing (Section 2). Given nonstationary time series $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$, we develop a nonlinear regression-based test of the null hypothesis that X_t is conditionally independent of Y_t given Z_t at all times t, with any desired leads and lags incorporated into these variables. Tests of conditional independence are used in many problems, such as variable selection, causal discovery, the validation of theory-based causal graphs, and the evaluation of predictor fairness; see Hardt et al. (2016).

II. Simulation-based Estimation (Section 3). Next, we introduce a simulation-based framework for estimating the unknown parameter in a parametric statistical model for a time series $X_{1:n}$. Such methods are desirable for models where the likelihood is analytically or computationally intractable, but simulating from the model is easy. For example, nonlinear non-Gaussian state-space models, discrete-time dynamical systems and ODEs with observational noise, and discretized Lévy-driven SDEs.

III. Prediction with Transformers (Section 4). Lastly, we develop a simulation-and-regression framework for training transformers for the tasks of forecasting and smoothing. Forecasting aims to characterize the future values of a stochastic process given the current information. It underpins monetary policy decisions in economics, portfolio optimization in finance, public health policy decisions during epidemics, and early warning systems for extreme weather events. Smoothing aims to characterize the past values of a latent stochastic process given the current information. Applications include estimating sentiment in the social sciences, stochastic volatility in finance, infectivity in epidemiology, and atmospheric variables in the Earth Sciences.

2 Conditional Independence Testing

2.1 Overview

In this part of the thesis, we introduce a framework for conditional independence testing with nonstationary time series. To the best of our knowledge, this is the first conditional independence testing framework for nonstationary time series outside of the linear-Gaussian setting. In particular, we use a theoretical framework for nonstationary time series based on the foundational work of Wu (2005); Zhou and Wu (2009); Dahlhaus et al. (2019). We propose a bootstrap-based testing procedure, which we justify with a distribution-uniform version of the strong Gaussian approximation from Mies and Steland (2023). We prove that our tests have Type I error control, *uniformly* over a large collection of null distributions. The main ideas are presented here, and we refer readers to the paper (Wieck-Sosa et al., 2025) for more details.

2.2 Setting

Let $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$ be time series taking values in $\mathbb{R}^{d_X}$, $\mathbb{R}^{d_Y}$, $\mathbb{R}^{d_Z}$, respectively, for some $n, d_X, d_Y, d_Z \in \mathbb{N}$. To simplify the notation, let these variables incorporate any desired leads and lags. The only requirement is that the conditioning set Z_t is known at time t. In this section, we present a test for the null hypothesis that

$$X_t \perp\!\!\!\perp Y_t \mid Z_t \text{ for all } t \in [n]. \tag{1}$$

We allow the time series to have nonlinear temporal dynamics and general forms of nonstationarity. The paper (Wieck-Sosa et al., 2025) includes the details about the asymptotic framework, the assumptions about the nonstationarity and decay of temporal dependence using the physical dependence measure of Wu (2005), and the rates at which the number of dimensions and time-offsets can grow with n.

2.3 Time-varying Regression and Error Processes

We can always decompose

$$X_t = f_t(Z_t) + \varepsilon_t, \quad Y_t = g_t(Z_t) + \xi_t, \tag{2}$$

where $f_t(z) = \mathbb{E}(X_t \mid Z_t = z)$ and $g_t(z) = \mathbb{E}(Y_t \mid Z_t = z)$ are the regression functions, which may vary over time. The error processes $\varepsilon_{1:n}$ and $\xi_{1:n}$ can also be nonstationary and temporally dependent. Denote the vectorized outer product of the errors at time t by

$$R_t = \mathrm{vec}(\varepsilon_t \xi_t^\top). \tag{3}$$

Next, let $\hat{f}_t$ and $\hat{g}_t$ be estimates of f_t and g_t, respectively, and let

$$\hat{\varepsilon}_t = X_t - \hat{f}_t(Z_t), \quad \hat{\xi}_t = Y_t - \hat{g}_t(Z_t), \tag{4}$$

be the corresponding residuals. Denote the vectorized outer product of these residuals at time t by

$$\hat{R}_t = \mathrm{vec}(\hat{\varepsilon}_t \hat{\xi}_t^\top). \tag{5}$$

Crucially, our testing framework can be used with *any regression estimator*. The required rates of convergence for the regression estimators are discussed in the paper (Wieck-Sosa et al., 2025). In our experiments, we use the sieve estimator from Ding and Zhou (2021), though there are many other regression estimators for nonstationary time series to choose from (Vogt, 2012; Zhang and Wu, 2015; Yousuf and Ng, 2021; Chen et al., 2022; Kurisu et al., 2025). Our test possesses a property known as *rate double robustness*, which means that we only require modest convergence rates for the products of the estimation errors, rather than for each estimation error individually.

2.4 Testing Procedure

Our proposed testing procedure can be viewed as an extension of the *generalized covariance measure* (GCM) test from Shah and Peters (2020) from the iid setting to the nonstationary time series setting, so we refer to it as the *dynamic generalized covariance measure* (dGCM) test. The core idea is that, under the null hypothesis of conditional independence from (1), the covariance of the errors from (2) will be zero at all times. Note that these covariances can be zero at all times, even under alternatives in which the corresponding conditional dependencies always hold. Consequently, we can only hope to have power against alternatives in which these covariances are non-zero for at least *some* times.

Roughly speaking, our test statistic uses the empirical covariances among the residuals from (4) to try to detect non-zero covariances among the errors from (2), for at least *some* times. To estimate the desired quantile of the test statistic, we use a bootstrap procedure justified by a *distribution-uniform* version of the strong Gaussian approximation from Mies and Steland (2023) applied to the process of error products R_t from (3). This is possible because, under the null hypothesis from (1), the error products R_t from (3) will have mean zero at all times t. The approximating Gaussian process has a time-varying covariance structure, which we estimate using a rolling-window approach; see the paper (Wieck-Sosa et al., 2025) for how to select the window size. We prove that the following procedure has Type I error control, uniformly over a large collection of null distributions.

Dynamic Generalized Covariance Measure (dGCM) Test.

- **Inputs:** Data $X_{1:n}$, $Y_{1:n}$, $Z_{1:n}$, significance level α, number of simulations s, $p \in [2, \infty]$ for norm in test statistic, and method for selecting the window size L.
- Regress X_t on Z_t, $t \in [n]$, and obtain residuals $\hat{\varepsilon}_t$, $t \in [n]$.
- Regress Y_t on Z_t, $t \in [n]$, and obtain residuals $\hat{\xi}_t$, $t \in [n]$.
- Obtain residual products $\hat{R}_t = \mathrm{vec}(\hat{\varepsilon}_t \hat{\xi}_t^\top)$, $t \in [n]$.
- Calculate test statistic on time series of residual products
$$T(\hat{R}_{1:n}) = \max_{j \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq j} \hat{R}_t \right\|_p .$$
- Select the window size L for covariance estimation using $\hat{R}_{1:n}$.
- For $t \in [n]$, obtain estimates of the time-varying covariances as
$$\hat{\Sigma}_t = \frac{1}{L} \left(\sum_{j=t-L+1}^{t} \hat{R}_j \right) \left(\sum_{j=t-L+1}^{t} \hat{R}_j \right)^\top .$$
- For each $r \in [s]$ and $t \in [n]$, simulate *independent* Gaussians
$$\tilde{R}_t^{(r)} \sim N(0, \hat{\Sigma}_t).$$
- For each $r \in [s]$, calculate test statistic on simulated Gaussian process
$$T(\tilde{R}_{1:n}^{(r)}) = \max_{j \in [n]} \left\| \frac{1}{\sqrt{n}} \sum_{t \leq j} \tilde{R}_t^{(r)} \right\|_p .$$
- Calculate the $1 - \alpha$ empirical quantile $\hat{q}_{1-\alpha}^{\mathrm{boot}}$ of the simulated test statistics $T(\tilde{R}_{1:n}^{(1)}), \ldots, T(\tilde{R}_{1:n}^{(s)})$.
- Reject null hypothesis from (1) at level α if $T(\hat{R}_{1:n}) > \hat{q}_{1-\alpha}^{\mathrm{boot}}$, else retain.

3 Simulation-based Estimation

3.1 Overview

In this part of the thesis, we develop a simulation-based estimation framework for parametric models of stochastic processes. Given an $\mathbb{R}^d$-valued time series $X_{1:n}$ generated by such a model, we aim to estimate the unknown parameter $\theta_0 \in \mathbb{R}^p$ for the model. Unfortunately, the likelihood function is intractable, but simulating from the model with different parameter values is straightforward.

Our core idea is to estimate θ_0 by tuning the parameter values so that a small number of *random features* of the simulated data closely match those of the observed data. Results from the "geometry from a time series" literature in nonlinear dynamics indicate that models with a p-dimensional parameter space can typically be identified using only $2p+1$ random features, regardless of the dimension of the observations. Our approach avoids both user-selected summary statistics and costly neural network-based procedures for learning them.

We introduce two estimators. First, a time-average estimator for processes that are stationary, or at least asymptotically mean stationary. Second, a rolling-window estimator for processes with general forms of nonstationarity. We prove that both estimators are consistent and asymptotically normal under nonrestrictive conditions. Specifically, we use a theoretical framework for nonstationary time series based on Wu (2005); Zhou and Wu (2009); Dahlhaus et al. (2019); Mies and Steland (2023). Our experiments suggest that simulation-based estimation can be made more automatic by matching random features, with little loss in statistical efficiency.

3.2 Notation and Setting

For real numbers $x, y \in \mathbb{R}$, denote $x \wedge y = \min(x, y)$ and $x \vee y = \max(x, y)$. For a vector $x \in \mathbb{R}^d$, denote the ℓ^p norm by $\|x\|_p$ and the Euclidean norm by $\|x\| = \|x\|_2$. For a random vector X, $\|X\|_{\mathcal{L}^q} = (\mathbb{E}\|X\|^q)^{1/q}$ denotes the $\mathcal{L}^q$ norm of the Euclidean norm. For any natural number $j \in \mathbb{N}$, denote $[j] = \{1, 2, \ldots, j\}$. For a sequence of random vectors X_t, $t = 1, \ldots, n$, each taking values in $\mathbb{R}^d$, we denote a subsequence as $X_{t_1:t_2} = (X_{t_1}, \ldots, X_{t_2})^\top$, which takes values in $\mathbb{R}^{(t_2 - t_1 + 1) \times d}$ for some $t_1, t_2 \in [n]$.

We observe a time series X_t^{obs}, $t = 1, \ldots, n$, taking values in $\mathbb{R}^d$ for some fixed dimension $d \in \mathbb{N}$. The observed time series is assumed to arise from a known generative model with an unknown p-dimensional parameter

$\theta_0 \in \Theta \subset \mathbb{R}^p$, $p \in \mathbb{N}$. Each value of $\theta \in \Theta$ determines a law of a stochastic process. For any value of θ, we can use the generative model to simulate $s \in \mathbb{N}$ realizations of a length n time series $(X_{1:n}^{(r)}(\theta))_{r\in[s]}$. When we do not need to refer to a particular realization, we drop the superscript.

3.3 Random Fourier Features

The central idea is that we should estimate the p-dimensional parameter θ_0 by matching the values of a small number of random features of the simulated and observed data. Consider $k = 2p+1$ randomly drawn functions $\varphi_1, \ldots, \varphi_k$ from a distribution D over a suitably nice function class $\mathcal{F}$. The k functions should be sampled independently of one another and of the data. Each function $\varphi_i : \mathbb{R}^{(m+1)\times d} \to \mathbb{R}$ takes as input a length $m+1$ subsequence of the d-dimensional time series, where m is chosen based on the dynamics of the generative model. For instance, if the model is an order m^* Markov process, then set $m = m^*$.

Let $x_1, \ldots, x_{m+1}$ be vectors in $\mathbb{R}^d$, and define $x = (x_1, \ldots, x_{m+1})^\top \in \mathbb{R}^{(m+1)\times d}$. We consider random Fourier features of the form

$$\varphi_i(x) = \cos\left(\sum_{j=1}^{m+1} \Omega_{i,j} \cdot x_j + \alpha_i\right), \tag{6}$$

where $\Omega_{i,j} \overset{\text{iid}}{\sim} N(0, I_d)$ and $\alpha_i \overset{\text{iid}}{\sim} U(-\pi,\pi)$ for $i = 1, \ldots, k$ and $j = 1, \ldots, m+1$. Denote all k functions by $\varphi = (\varphi_1, \ldots, \varphi_k)$, so that $\varphi : \mathbb{R}^{(m+1)\times d} \to \mathbb{R}^k$. Define the i-th random feature of the observed and r-th simulated time series at time $t = m+1, \ldots, n$ as

$$f_{t,i}^{\text{obs}} = \varphi_i(X_{t-m:t}^{\text{obs}}), \quad f_{t,i}^{(r)}(\theta) = \varphi_i\left(X_{t-m:t}^{(r)}(\theta)\right).$$

Write all k random features at time t as

$$f_t^{\text{obs}} = \left(f_{t,1}^{\text{obs}}, \ldots, f_{t,k}^{\text{obs}}\right), \quad f_t^{(r)}(\theta) = \left(f_{t,1}^{(r)}(\theta), \ldots, f_{t,k}^{(r)}(\theta)\right). \tag{7}$$

3.4 Estimators

Time-average estimator. For processes that are asymptotically mean stationary, it suffices to consider the time-average of $k = 2p+1$ random features. Define the observed and simulated time-averages of the random features by

$$F^{\text{obs}} = \frac{1}{n-m}\sum_{t=m+1}^{n} f_t^{\text{obs}}, \quad \bar{F}^{\text{sim}}(\theta) = \frac{1}{n-m}\sum_{t=m+1}^{n} \bar{f}_t^{\text{sim}}(\theta), \tag{8}$$

where $\bar{f}_t^{\text{sim}}(\theta) = \frac{1}{s}\sum_{r=1}^{s} f_t^{(r)}(\theta)$. The time-average estimator is given by

$$\hat{\theta}^{\text{TA}} = \operatorname*{argmin}_{\theta\in\Theta} \hat{Q}_n^{\text{TA}}(\theta), \tag{9}$$

where the objective function is defined as

$$\hat{Q}_n^{\text{TA}}(\theta) = \left\| F^{\text{obs}} - \bar{F}^{\text{sim}}(\theta) \right\|^2. \tag{10}$$

Rolling-window estimator. For many nonstationary processes, a different approach is required because the average of the time-varying means of the random features over time may not converge to a limiting mean. Even when it does converge, the limiting mean may not uniquely identify each θ. Thus, we minimize the time-average of the squared distances between rolling-window averages of $k = 2p + 1$ random features.

Define the observed and simulated rolling-window random features at time t by

$$F_t^{\text{obs}} = \frac{1}{w \wedge t} \sum_{j=(t-w)\vee 1}^{t} f_j^{\text{obs}}, \quad \bar{F}_t^{\text{sim}}(\theta) = \frac{1}{w \wedge t} \sum_{j=(t-w)\vee 1}^{t} \bar{f}_j^{\text{sim}}(\theta), \tag{11}$$

where $\bar{f}_t^{\text{sim}}(\theta) = \frac{1}{s}\sum_{r=1}^{s} f_t^{(r)}(\theta)$ and $w \in \mathbb{N}$ is the window size. The rolling-window estimator is given by

$$\hat{\theta}^{\text{RW}} = \operatorname*{argmin}_{\theta\in\Theta} \hat{Q}_n^{\text{RW}}(\theta), \tag{12}$$

where the objective function is defined as

$$\begin{aligned}
\hat{Q}_n^{\text{RW}}(\theta) &= \frac{1}{n-m} \sum_{t=m+\tau+L}^{n} \left[\left\| F_t^{\text{obs}} - \bar{F}_t^{\text{sim}}(\theta) \right\|^2 + K_t(\theta) \right], \\
K_t(\theta) &= 2 \left(F_{t-L}^{\text{obs}} - \bar{F}_{t-L}^{\text{sim}}(\theta) \right)^\top \left(\left[f_t^{\text{obs}} - \bar{f}_t^{\text{sim}}(\theta) \right] - \left[F_{t-L}^{\text{obs}} - \bar{F}_{t-L}^{\text{sim}}(\theta) \right] \right),
\end{aligned}$$

where $\tau \in \mathbb{N}$ is an initial time-offset and $L \in \mathbb{N}$ is a lag.

In practice, we select the window size, offset, and lag in a similar manner as Mies (2023). The window size w_n is selected as the integer within $[n^{\frac{1}{2}}, n^{\frac{3}{4}}]$ which minimizes the sum of squared distances $\sum_{t=m+1+L}^{n} \left\| F_{t-L}^{\text{obs}} - f_t^{\text{obs}} \right\|^2$, and we set the offset as $\tau_n = w_n$. We let the lag L_n grow slowly with n at the polylogarithmic rate $\lceil \frac{1}{10} \log(n)^2 \rceil$.

Extensions. Several extensions are possible. First, both estimators can be generalized by replacing the Euclidean norm with a weighted norm, using a weight matrix given by the inverse of a suitable estimator of the long-run covariance matrix; see Gourieroux and Monfort (1996). Second, the sample averages used in both estimators can be replaced with different mean estimators. Third, one may extend both estimators to allow for early stopping, so that only a fraction of the time series is used.

3.5 Examples

To illustrate the ideas, let us consider a few examples. The paper contains several more examples. We optimize the objective functions using the differential evolution solver `differential_evolution` from the `scipy.optimize` module in the SciPy Python library, although our experiments indicate that other optimizers perform comparably. For the SIR example, we used the Runge–Kutta 5(4) solver `RK45` from the `scipy.integrate` module in SciPy. Each density plot is constructed from $1,000$ independent estimates of the p-dimensional parameter using $2p+1$ random Fourier features and 10 simulations per parameter value.

Example 3.1 (Gaussian)**.** For $t = 1, \ldots, n$, we observe

$$X_t = \mu + \epsilon_t,$$

where $\epsilon_t \overset{\text{iid}}{\sim} N(0,1)$. The unknown parameter is μ, so we use $2p + 1 = 3$ random features.

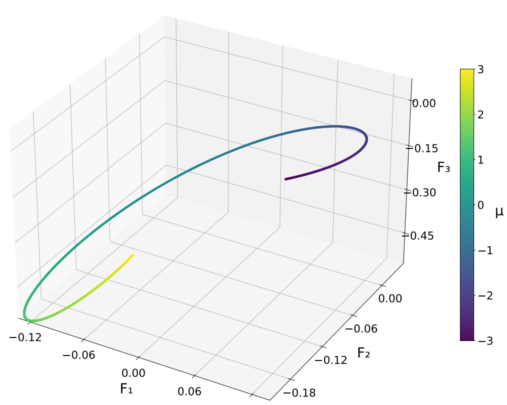
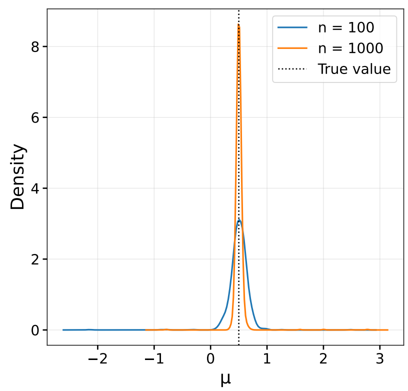


Figure 1: Three random features averaged over $n = 1000$ times and $s = 10$ simulations, for each parameter value on a grid in $[-3, 3]$. Color denotes μ.

We aim to estimate

$$\mu_0 = 0.5,$$

using the time-average estimator.



Example 3.2 (Logistic Map)**.** For $t = 1, \ldots, n$, define the state by

$$Z_t = \rho Z_{t-1}(1 - Z_{t-1}).$$

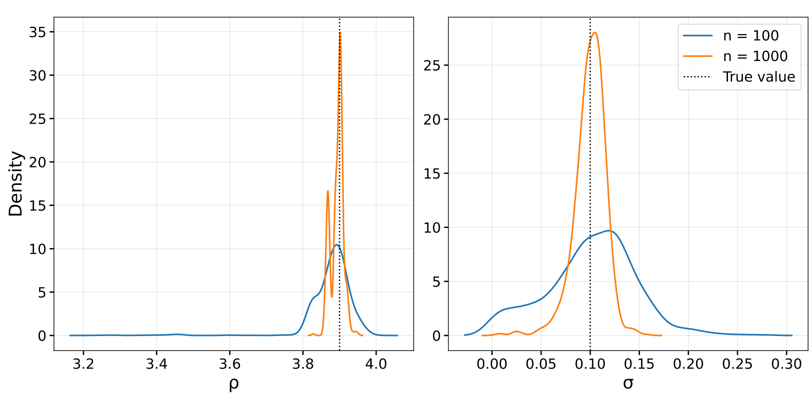
We observe

$$X_t = Z_t + \sigma \epsilon_t,$$

where $\epsilon_t \overset{\text{iid}}{\sim} N(0, 1)$ and the unknown initial value Z_0 is sampled from a $U[0, 1]$ distribution, so that it is fixed before running the procedure. Our goal is to estimate

$$\rho_0 = 3.9, \quad \sigma_0 = 0.1,$$

using the time-average estimator. We emphasize that the logistic map is in a chaotic regime when $\rho = 3.9$; see Devaney (2018) for more information.



Example 3.3 (SIR model)**.** For time horizon $T = 50$, let $(S_v, I_v, R_v)_{v\in[0,T]}$, be defined by the susceptible-infected-recovered (SIR) system of ODEs

$$\begin{aligned}\frac{\mathrm{d}S_v}{\mathrm{d}v} &= -\frac{\beta}{N} S_v\, I_v,\\ \frac{\mathrm{d}I_v}{\mathrm{d}v} &= \frac{\beta}{N} S_v\, I_v - \gamma I_v,\\ \frac{\mathrm{d}R_v}{\mathrm{d}v} &= \gamma I_v,\end{aligned}$$

where S_v/N, I_v/N, and R_v/N denote the fractions of the population that are susceptible, infected, and recovered (or removed), respectively, $\beta > 0$ is the transmission rate, and $\gamma > 0$ is the recovery rate. Let the known initial values be $S_0 = 980{,}000$, $I_0 = 20{,}000$, $R_0 = 0$, where the number of individuals is $N = S_0 + I_0 + R_0 = 1{,}000{,}000$.

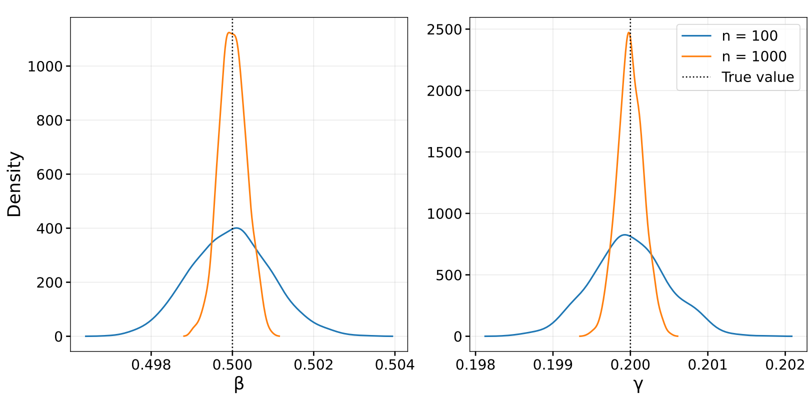
For each $t = 1, \ldots, n$, we test $\mathrm{TE}_t = \lfloor N/100 \rfloor$ people with equal probability using a test with sensitivity $\mathrm{SE}_t = 0.95$ and specificity $\mathrm{SP}_t = 0.99$. We observe the proportion of positive tests

$$X_t = Z_t/\mathrm{TE}_t, \quad Z_t \sim \mathrm{Binomial}\left(\mathrm{TE}_t, \pi_t\right),$$

where $\pi_t = \mathrm{SE}_t I_{Tt/n}/N + (1 - \mathrm{SP}_t)\left(1 - I_{Tt/n}/N\right)$. This model allows the number of tests TE_t, sensitivity SE_t, and specificity SP_t to change with t, although we keep them constant for simplicity. We aim to estimate

$$\beta_0 = 0.5, \quad \gamma_0 = 0.2,$$

using the rolling-window estimator. Note that the bounds for the optimizer were set to $[0.01, 1]$ for β and $[0.01, 0.5]$ for γ.



4 Prediction with Transformers

4.1 Overview

Drawing inspiration from the remarkable success of transformer-based foundation models for language and vision tasks, many researchers are now seeking to develop foundation models for time series using transformers (Vaswani et al., 2017). *Time series foundation models* (TSFMs) are large models, with millions or even billions of parameters, that are pretrained on massive collections of real-world and synthetic time series. There has recently been a growing emphasis on training TSFMs using synthetic time series (Ansari et al., 2024, 2025; Liu et al., 2025). In fact, many TSFMs are trained *exclusively* on synthetic time series (Dooley et al., 2023; Bhethanabhotla et al., 2024; Verdenius et al., 2024; Taga et al., 2025; Oreshkin et al., 2026). The synthetic time series are often generated by *parametric* time series models with a variety of different parameter values. Many TSFMs are also trained on convex combinations of time series realizations; see TSMixup from Ansari et al. (2024). These rapid developments raise fundamental questions about how transformers trained on synthetic time series data are able to learn useful representations of temporal dynamics, especially in nonstationary settings.

Additionally, we take inspiration from the *simulation-and-regression* methods of Longstaff and Schwartz (2001) and Moussa et al. (2026). The *least squares Monte Carlo* (LSMC) method from Longstaff and Schwartz (2001) was originally developed for pricing American options. For the purposes of this discussion, we will gloss over the details involving finance. In the LSMC algorithm, simulation is used to estimate the regression function $f(y) = \mathbb{E}(X \mid Y = y)$ for two random variables X, Y by drawing $X^{(r)}$, $Y^{(r)}$, $r = 1, \ldots, s$, and minimizing the squared loss

$$\frac{1}{s} \sum_{r=1}^{s} (X^{(r)} - f(Y^{(r)}))^2,$$

over some function class to obtain an estimate $\hat{f}$ of the regression function f. Similarly, the *extremum Monte Carlo* (XMC) method from Moussa et al. (2026) uses nonparametric regression, such as random forests and XGBoost, to perform forecasting and smoothing for state-space models when the parameter is assumed to be known. In practice, the parameter is typically estimated and treated as known. Consequently, the regression estimators in the XMC method minimize an expected loss, where the expectation is with respect to a single distribution in the model.

Motivated by these topics, we propose to study time series prediction with transformers using a simulation-and-regression approach that accounts for parameter uncertainty. We focus on using transformers for two concrete tasks: (1) forecasting future observations and latent states, and (2) smoothing past latent states; see Section 4.2. We will investigate different strategies for training transformers for these tasks, given a fixed simulation budget. Our goals are to: (1) understand the statistical foundations of simulation-based training approaches for transformers, and (2) identify principles that enable transformers to achieve satisfactory performance on time series prediction tasks. The proposed work is presented in Section 4.3.

4.2 State-space Modeling

We give a brief overview of forecasting and smoothing for state-space models. We emphasize that our proposed work on transformer-based methods for time series is not restricted to state-space models. Nevertheless, the following definitions and examples will be helpful to have in mind for the discussions in Section 4.3.

State-space models decompose observed time series $Y_{1:t}$ into latent states $X_{1:t}$, control inputs $Z_{1:t}$, and noise. Consider the (possibly nonstationary) state-space model for an $\mathbb{R}^{d_Y}$-valued observed process $Y_{1:n}$ and an $\mathbb{R}^{d_X}$-valued latent state process $X_{1:n}$. For times $t = 1, \ldots, n$, let

$$\begin{aligned} Y_t &= M_t(X_t, Z_t, \varepsilon_t^Y, \theta), \qquad & (\varepsilon_t^X, \varepsilon_t^Y) &\sim P_\theta^{\varepsilon_t^X, \varepsilon_t^Y}, \\ X_t &= S_t(X_{t-1}, Z_t, \varepsilon_t^X, \theta), \qquad & X_0 &\sim P_\theta^{X_0}, \end{aligned} \tag{13}$$

where M_t is the measurement function at time t, S_t is the state transition function at time t, and $\theta \in \Theta \subset \mathbb{R}^p$ is a p-dimensional parameter.

To give a sense of the expressiveness of state-space models, we present some simple examples. First, consider an autoregressive process (with no controls) observed through noise

$$\begin{aligned} Y_t &= X_t + \varepsilon_t^Y, \qquad & (\varepsilon_t^X, \varepsilon_t^Y) &\sim N(\mathbf{0}, \mathbf{\Sigma}), \\ X_t &= \phi X_{t-1} + \varepsilon_t^X, \qquad & X_0 &\sim N(0, \sigma_1^2), \end{aligned} \tag{14}$$

where

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_X^2 & \rho\,\sigma_X \sigma_Y \\ \rho\,\sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix},$$

for some $\phi \in (-1, 1)$, $\rho \in [0, 1]$, and $\sigma_1, \sigma_X, \sigma_Y > 0$. Second, consider a discrete-valued, non-Gaussian process with $\mathbb{R}^{d_Z}$-valued controls $Z_{1:n}$

$$\begin{aligned} Y_t &\sim \text{Poisson}(X_t), \\ \log(X_t) &= \beta\alpha_t + \beta_Z^\top Z_t + \mu, \\ \alpha_t &= \phi\alpha_{t-1} + \epsilon_t, \end{aligned} \tag{15}$$

where $\epsilon_t \sim t_5$, $\alpha_0 \sim t_5$, $\phi \in (-1, 1)$, $\beta \in \mathbb{R}$, $\beta_Z \in \mathbb{R}^{d_Z}$, and $\mu \in \mathbb{R}$.

In the context of state-space modeling, *smoothing* is the task of estimating the past latent states $X_{1:t}$ using the information up to time t. This can be done via simulation-based estimates of the regression functions $\mathbb{E}(X_j \mid Y_{1:t} = y_{1:t}, Z_{1:t} = z_{1:t})$ for $j = 1, \ldots, t$. *Forecasting* is the task of predicting the future observations or states multiple time-steps ahead. In the state-space setting, a future path of the control inputs must be specified, either deterministically or stochastically, or the model must be specified so that no controls are used. Given the observations $Y_{1:t}$ and (optionally) control inputs $Z_{1:t+h}$, we aim to make point forecasts via simulation-based estimates of $\mathbb{E}(X_{t+j} \mid Y_{1:t} = y_{1:t}, Z_{1:t+j} = z_{1:t+j})$ or $\mathbb{E}(Y_{t+j} \mid Y_{1:t} = y_{1:t}, Z_{1:t+j} = z_{1:t+j})$ for $j = 1, \ldots, h$. Similar ideas apply for estimating the conditional quantiles or conditional distributions.

4.3 Proposed Work

We propose a transformer-based simulation-and-regression framework that incorporates parameter uncertainty. We suggest minimizing an expected sequence-to-sequence loss, where the expectation is with respect to a mixture of distributions in a model class. The mixture of distributions is based on a Gaussian approximation to the sampling distribution of the parameter estimator.

Our approach can be viewed as a frequentist, likelihood-free analogue of the Laplace approximation. Specifically, when using the Laplace approximation to the posterior distribution to obtain an approximate posterior predictive distribution. The Laplace approximation uses a Gaussian distribution centered at the maximum a posteriori (MAP) estimate with covariance given by the inverse observed Fisher information, whereas our approximation uses a Gaussian distribution centered at the estimated parameter with covariance given by an estimate of the asymptotic covariance matrix. The Bernstein–von Mises theorem is used to justify the Laplace approximation, whereas we use asymptotic theory for our (likelihood-free) parameter estimator to justify our approach.

In contrast to the simulation-and-regression methods from Longstaff and Schwartz (2001) and Moussa et al. (2026) discussed in Section 4.1, our proposed method (1) minimizes the expected sequence-to-sequence loss, where the expectation is with respect to a mixture of distributions in the model class, and (2) is designed for general purpose time series prediction tasks. Our proposed method can be used, in particular, for forecasting and smoothing in state-space models. In the context of smoothing and multi-horizon forecasting, our proposed method only requires training one model, as opposed to separate models for each time.

The previously described approach can be generalized by minimizing the conditional value-at-risk (CVaR), i.e. the expectation of the highest α-fraction of the losses, $\alpha \in (0, 1]$, where the expectation is with respect to the same mixture of distributions in the model class. This approach is commonly used in distributionally robust optimization (Rockafellar and Uryasev, 2000; Takeda and Kanamori, 2009; Shalev-Shwartz and Wexler, 2016; Soma and Yoshida, 2020; Levy et al., 2020). Note that, when $\alpha = 1$, this coincides with the previously described method.

We aim to show that the estimator obtained by the proposed procedure is consistent as the sample size n and number of simulations s tend to infinity, with convergence rate determined by the parameter estimator and sequence-to-sequence regression estimator. We will study the finite-sample performance of the proposed procedures and compare with existing approaches. If time allows, we will also consider text-numeric time series observations.

The proposed methods are presented below. To simplify matters, we present the proposed methods in the context of smoothing. Specifically, smoothing past states via estimates of their conditional expectations given the current information. Similar ideas apply for conditional quantile estimation, conditional distribution estimation, and forecasting of future observations and latent states.

The algorithm utilizes an estimator $\hat{\theta}$ of the true parameter θ_0, and it is assumed to be consistent and asymptotically normal. For instance, we can use the random feature estimators from Section 3. The control inputs are treated as fixed, and are absorbed into the definitions of the measurement function and state transition function from (13). For the cross-validation scheme, we introduce an aggregation function $\mathrm{Agg}(\cdot)$ for more flexibility. For example, $\mathrm{Agg}(\cdot)$ can be taken to be the average over the K folds, or just the error in the validation fold if using an 80/20 train/validation split.

Transformer Simulation-and-Regression (Minimize Average Loss).

- **Inputs:** Observations $Y_{1:n}^{\mathrm{obs}}$, state-space model (13), number of simulations s, number of folds K for cross-validation, proportion of simulations in each fold $c_1, \dots, c_K$, cross-validation error aggregation function $\mathrm{Agg}(\cdot)$, estimator $\hat{\theta}$, and estimation method for asymptotic covariance of $\hat{\theta}$.

- Obtain estimate $\hat{\theta}$ of the true parameter θ_0.
- Obtain estimate $\hat{\Sigma}$ of the asymptotic covariance Σ of estimator $\hat{\theta}$.
- For each $r \in [s]$, draw $\theta^{(r)} \sim N(\hat{\theta}, \hat{\Sigma}/n)$ and simulate from the model (13),
$$X_{1:n}^{(r)}, Y_{1:n}^{(r)} \sim P_{n,\theta^{(r)}}.$$
- Denote the r-th sequence-to-sequence loss by
$$\ell_r(f) = \sum_{t \in [n]} (X_t^{(r)} - [f(Y_{1:n}^{(r)})]_t)^2.$$
- Partition $[s]$ into K folds $F_1, \dots, F_K$, where,
$$F_i = \{r \in [s] : \sum_{j=1}^{i-1} c_j s < r \leq \sum_{j=1}^{i} c_j s\}, \quad i = 1, \dots, K.$$
- For each $k \in [K]$ and each value of the tuning parameter $\lambda \in \Lambda$, perform sequence-to-sequence regression to obtain the minimizer $\hat{f}_\lambda^{-k}$ of the average loss
$$\frac{1}{|F_{-k}|} \sum_{r \in F_{-k}} \ell_r(f),$$
over the function space of transformers, where $F_{-k} = \bigcup_{j \in [K] \setminus \{k\}} F_j$.
- Record the corresponding error on the k-th validation set
$$e_k(\lambda) = \frac{1}{|F_k|} \sum_{r \in F_k} \ell_r(\hat{f}_\lambda^{-k}).$$
- Compute the aggregated error over the K folds
$$\mathrm{CV}(\lambda) = \mathrm{Agg}(e_1(\lambda), \dots, e_K(\lambda)).$$
- Find the value λ^* of the tuning parameter λ that minimizes $\mathrm{CV}(\lambda)$.

- Obtain the estimate $\hat{f}$ by minimizing the average loss using all s simulations with the selected tuning parameter λ^* to

$$\frac{1}{s}\sum_{r\in[s]}\ell_r(f).$$

- Apply the estimated function $\hat{f}$ to predict the latent states

$$\hat{X}_{1:n} = \hat{f}(Y^{\text{obs}}_{1:n}).$$

Transformer Simulation-and-Regression (Minimize CVaR Loss).

- **Inputs:** Observations $Y^{\text{obs}}_{1:n}$, state-space model (13), number of simulations s, number of folds K for cross-validation, proportion of simulations in each fold $c_1, \dots, c_K$, cross-validation error aggregation function $\text{Agg}(\cdot)$, estimator $\hat{\theta}$, estimation method for asymptotic covariance of $\hat{\theta}$, and fraction $\alpha \in (0, 1]$ for conditional value-at-risk objective.
- Obtain estimate $\hat{\theta}$ of the true parameter θ_0.
- Obtain estimate $\hat{\Sigma}$ of the asymptotic covariance Σ of estimator $\hat{\theta}$.
- For each $r \in [s]$, draw $\theta^{(r)} \sim N(\hat{\theta}, \hat{\Sigma}/n)$ and simulate from the model (13),

$$X^{(r)}_{1:n}, Y^{(r)}_{1:n} \sim P_{n,\theta^{(r)}}.$$

- Denote the r-th sequence-to-sequence loss by

$$\ell_r(f) = \sum_{t\in[n]} (X^{(r)}_t - [f(Y^{(r)}_{1:n})]_t)^2.$$

- Partition $[s]$ into K folds $F_1, \dots, F_K$, where,

$$F_i = \{r \in [s] : \sum_{j=1}^{i-1} c_j s < r \le \sum_{j=1}^{i} c_j s\}, \quad i = 1, \dots, K.$$

- For each $k \in [K]$ and each value of the tuning parameter $\lambda \in \Lambda$, perform sequence-to-sequence regression to obtain the minimizer $\hat{f}^{-k}_\lambda$ of the empirical conditional value-at-risk

$$\inf_{\eta\in\mathbb{R}} \left(\eta + \frac{1}{\alpha|F_{-k}|}\sum_{r\in F_{-k}} (\ell_r(f) - \eta)_+\right),$$

over the function space of transformers, where $F_{-k} = \bigcup_{j\in[K]\setminus\{k\}} F_j$.

- Record the corresponding error on the k-th validation set

$$e_k(\lambda) = \inf_{\eta \in \mathbb{R}} \left(\eta + \frac{1}{\alpha |F_k|} \sum_{r \in F_k} \left(\ell_r(\hat{f}_\lambda^{-k}) - \eta \right)_+ \right).$$

- Compute the aggregated error over the K folds

$$\mathrm{CV}(\lambda) = \mathrm{Agg}(e_1(\lambda), \ldots, e_K(\lambda)).$$

- Find the value λ^* of the tuning parameter λ that minimizes $\mathrm{CV}(\lambda)$.

- Obtain the estimate $\hat{f}$ by minimizing the conditional value-at-risk objective using all s simulations with the selected tuning parameter λ^*

$$\inf_{\eta \in \mathbb{R}} \left(\eta + \frac{1}{\alpha s} \sum_{r \in [s]} (\ell_r(f) - \eta)_+ \right).$$

- Apply the estimated function $\hat{f}$ to predict the latent states

$$\hat{X}_{1:n} = \hat{f}(Y_{1:n}^{\mathrm{obs}}).$$

5 Timeline

Step 1a: Proposal. I plan to propose my thesis in April, 2026.

Step 1b: Revise and Submit. Concurrently, I plan to revise the conditional independence testing paper corresponding to Section 2, which has received a revise decision from the journal, and to submit the simulation-based parameter estimation paper corresponding to Section 3.

Step 2: Prediction with Transformers (Section 4). I plan to work on this topic from April 2026 to March 2027.

Step 3: Defense. I plan to defend my thesis before the spring defense deadline set by the Department, April 15, 2027, and submit the paperwork by May 1, 2027.

References

Ansari, A. F., Shchur, O., Küken, J., Auer, A., Han, B., Mercado, P., Rangapuram, S. S., et al. (2025). Chronos-2: From univariate to universal forecasting. arXiv preprint arXiv:2510.15821.

Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., Shchur, O., et al. (2024). Chronos: Learning the language of time series. arXiv preprint arXiv:2403.07815.

Bhethanabhotla, S. K., Swelam, O., Siems, J., Salinas, D., and Hutter, F. (2024). Mamba4cast: Efficient zero-shot time series forecasting with state space models. arXiv preprint arXiv:2410.09385.

Chen, L., Smetanina, E., and Wu, W. B. (2022). Estimation of nonstationary nonparametric regression model with multiplicative structure. *The Econometrics Journal*, 25(1):176–214.

Dahlhaus, R., Richter, S., and Wu, W. B. (2019). Towards a general theory for nonlinear locally stationary processes. *Bernoulli*, 25(2):1013–1044.

Devaney, R. L. (2018). *A first course in chaotic dynamical systems: theory and experiment*. CRC Press.

Ding, X. and Zhou, Z. (2021). Simultaneous sieve inference for time-inhomogeneous nonlinear time series regression. arXiv preprint arXiv:2112.08545.

Dooley, S., Khurana, G. S., Mohapatra, C., Naidu, S. V., White, C., et al. (2023). Forecastpfn: Synthetically-trained zero-shot forecasting. *Advances in Neural Information Processing Systems*, 36:2403–2426.

Gourieroux, C. and Monfort, A. (1996). *Simulation-based econometric methods*. Oxford University Press, Oxford, England.

Hardt, M., Price, E., and Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems*, 29:3323–3331.

Kurisu, D., Fukami, R., and Koike, Y. (2025). Adaptive deep learning for nonlinear time series models. *Bernoulli*, 31(1):240–270.

Levy, D., Carmon, Y., Duchi, J. C., and Sidford, A. (2020). Large-scale methods for distributionally robust optimization. *Advances in Neural Information Processing Systems*, 33:8847–8860.

Liu, X., Aksu, T., Liu, J., Wen, Q., Liang, Y., Xiong, C., Savarese, S., Sahoo, D., Li, J., and Liu, C. (2025). Empowering time series analysis with synthetic data: A survey and outlook in the era of foundation models. arXiv preprint arXiv:2503.11411.

Longstaff, F. A. and Schwartz, E. S. (2001). Valuing american options by simulation: A simple least-squares approach. *The Review of Financial Studies*, 14(1):113–147.

Mies, F. (2023). Functional estimation and change detection for nonstationary time series. *Journal of the American Statistical Association*, 118(542):1011–1022.

Mies, F. and Steland, A. (2023). Sequential gaussian approximation for nonstationary time series in high dimensions. *Bernoulli*, 29(4):3114–3140.

Moussa, K., Blasques, F., and Koopman, S. J. (2026). Extremum monte carlo filters: signal extraction via simulation and regression. *Journal of Business and Economic Statistics*, pages 1–13.

Oreshkin, B. N., Jauhari, M., Selvam, R. K., Wolff, M., Pan, W., Ramasubramanian, S., Olivares, K. G., et al. (2026). Zero-shot forecasting by simulation alone. arXiv preprint arXiv:2601.00970.

Rockafellar, R. T. and Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2:21–42.

Shah, R. D. and Peters, J. (2020). The hardness of conditional independence testing and the generalised covariance measure. *Annals of Statistics*, 48(3):1514–1538.

Shalev-Shwartz, S. and Wexler, Y. (2016). Minimizing the maximal loss: How and why. *International Conference on Machine Learning*, pages 793–801.

Soma, T. and Yoshida, Y. (2020). Statistical learning with conditional value at risk. arXiv preprint arXiv:2002.05826.

Taga, E. O., Ildiz, M. E., and Oymak, S. (2025). Timepfn: Effective multivariate time series forecasting with synthetic data. *Proceedings of the AAAI conference on artificial intelligence*, 39(19):20761–20769.

Takeda, A. and Kanamori, T. (2009). A robust approach based on conditional value-at-risk measure to statistical learning problems. *European Journal of Operational Research*, 198(1):287–296.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Łukasz Kaiser, and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30:3323–3331.

Verdenius, S., Zerio, A., and Wang, R. L. (2024). Lat-pfn: A joint embedding predictive architecture for in-context time-series forecasting. arXiv preprint arXiv:2405.10093.

Vogt, M. (2012). Nonparametric regression for locally stationary time series. *Annals of Statistics*, 50(5):2601–2633.

Wieck-Sosa, M., Haddad, M. F. C., and Ramdas, A. (2025). Conditional independence testing with a single realization of a multivariate nonstationary nonlinear time series. arXiv preprint arXiv:2504.21647.

Wu, W. B. (2005). Nonlinear system theory: another look at dependence. *Proceedings of the National Academy of Sciences*, 102(40):14150–14154.

Yousuf, K. and Ng, S. (2021). Boosting high dimensional predictive regressions with time varying parameters. *Journal of Econometrics*, 224(1):60–87.

Zhang, T. and Wu, W. B. (2015). Time-varying nonlinear regression models: nonparametric estimation and model selection. *Annals of Statistics*, 43(2):741–768.

Zhou, Z. and Wu, W. B. (2009). Local linear quantile estimation for nonstationary time series. *The Annals of Statistics*, 37(5B):2696–2729.